

Radioastronomy

Fortgeschrittenenpraktikum

HS1 Radioastronomy

Léonie Lanz

Nico Kalogerakis

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1. Introduction

The experiment HS1-Radioastronomy is there to give an overview for operating a radio telescope and to introduce the prerequisites such as characterising and calibrating the telescope before carrying out the experiments. The main experiment to be conducted is the observation of neutral hydrogen in the Milky Way to determine the rotational velocity in order to derive the mass distribution.

1.1. Radioastronomy

Radio astronomy is the study of electromagnetic radiation in the radio wavelength range, typically from a few MHz to several hundred GHz. Compared to optical astronomy, radio waves are much less affected by interstellar dust and gas, which allows astronomers to observe cold and dense regions of space that are often hidden in the optical regime. Radio telescopes usually consist of large parabolic antennas that collect incoming radiation and direct it to a receiver. The receiver amplifies the weak signal and converts it into an electrical voltage, from which the intensity, spectral distribution, and temperature of the observed source can be determined.

A central quantity in radio astronomy is the brightness temperature. In the Rayleigh-Jeans regime, which is valid for most radio astronomical observations, the spectral intensity is approximately

$$I_\nu \approx \frac{2k_B T_B \nu^2}{c^2} = \frac{2k_B T_B}{\lambda^2},$$

where I_ν is the spectral intensity, k_B is the Boltzmann constant, T_B is the brightness temperature, ν is the frequency, and c is the speed of light. This relation shows that the observed radio signal is directly linked to the temperature of the emitting medium [1]. This makes radio astronomy a powerful method for studying cold molecular clouds, ionized gas regions, and neutral hydrogen structures in the Milky Way and beyond.

One of the most important spectral lines in radio astronomy is the 21 cm line emitted by neutral hydrogen.

1.2. The 21 cm Hydrogen Line

The hydrogen atom consists of a proton and an electron. In the ground state, the proton and electron spins can be oriented either parallel or antiparallel to each other. This gives rise to a hyperfine splitting of the ground state: the antiparallel configuration has a slightly lower energy than the parallel configuration. A transition from the higher-energy state to the lower-energy state leads to the emission of a photon with a frequency of approximately

$$\nu \approx 1420.4 \text{ MHz},$$

which corresponds to a wavelength of

$$\lambda \approx 21.1 \text{ cm}.$$

This transition is therefore called the 21 cm line.

This line is of great importance in astronomy because hydrogen is the most abundant element in the interstellar medium. Neutral hydrogen emits strongly at 21 cm, allowing astronomers to map the distribution of gas in galaxies. The 21 cm line is especially useful because it is only weakly absorbed by interstellar dust, so it can reveal regions that are difficult or impossible to observe in the optical range.

The 21 cm line also enables the measurement of Doppler shifts. If the hydrogen gas is moving relative to the observer, the frequency of the emission is shifted. From this shift, the radial velocity of the gas can be determined. This makes it possible to study galactic rotation, gas dynamics, and the mass distribution of galaxies. For this reason, the 21 cm line is one of the most important tools in modern radio astronomy and is fundamental for studying neutral hydrogen in the universe.

2. Observation and Evaluation

2.1. The KRT3 telescope

2.2. Weather Conditions

The experiment was carried out on 07.09.2026, starting at around 12:35. The weather was stable throughout the day with partial cloud coverage and temperatures around 19°, a relative humidity of 60% and a windspeed of 1 m s⁻¹.

2.3. Calibration

2.3.1. Determination of the system temperature

As the receiver receives radiation not only from the source T_S but also from the environment T_h . In addition to those two external sources of radiation, the antenna also picks up on the radiation it's own system emits T_{sys} . This means the measured power of the receiver, represented by the antenna temperature, can be broken down the following way:

$$T_A = T_S + T_h + T_{sys} \quad (1)$$

It is difficult to separate the pure system temperature and the background radiation, so these temperatures are usually combined into an effective system temperature:

$$T'_{sys} = T_{sys} + T_h \quad (2)$$

The effective system temperature now includes all unwanted contributions. This means that to know the contribution of a source, one must know the effective system temperature. Determining the effective system temperature is calibrating the telescope by doing a 'On-Off' measurement through measuring a source with a known flux density and then measuring a piece of empty sky. The KRT3 however, can only measure the sun as a source with high enough flux density but the sun is not a constant source which is why a noise generator needs to be used.

The noise generator injects a signal directly into the antenna horn. The calibration temperature $T_{cal} = 170$ K is the expected antenna temperature that is observed additionally when the noise generator is switched on.

When assuming the sun to be a point-like source ?? can be used in the form of:

$$T_b \Omega_s = T_A \Omega_A \quad (3)$$

with T_b being the brightness temperature, Ω_s being the solid angle of the sun, T_A being the antenna temperature and Ω_A being the solid angle of the antenna main lobe.

The goal is measuring the antenna temperature of the sun and converting it into a flux density. For this we need the following identities: the antenna theorem $A_e \Omega_A = \lambda^2$, the Rayleigh-Jeans law $B_\nu = 2kT_b/\lambda^2$, the flux density $S_\nu = B_\nu \Omega_s$ and the aperture efficiency $\eta = A_e/A_g = 0.5$ with A_g being the geometric antenna surface assumed to be a circle.

$$T_b \Omega_s = T_A \Omega_A \quad (4)$$

$$\Omega_A = \frac{\lambda^2}{A_e}, \Omega_s = \frac{S_\nu}{B_\nu}, B_\nu = \frac{2k_B T_b}{\lambda^2} \quad (5)$$

$$S_\nu = \frac{T_A \cdot 2k_B}{A_e} = f(T_A, A_e) \quad (6)$$

$$A_e = \pi r^2 \cdot \eta = 1,5^2 \pi \cdot 0,5 = \pi 1,125 \text{ m}^2 \quad (7)$$

$$\Rightarrow T_A = \frac{S_\nu A_e}{2k_B} = \frac{S_\nu \pi 1,125 \text{ m}^2}{2k_B} \quad (8)$$

2.3.2. Background Radiation

The determination of the background radiation is done at the calibration position where $T_A \approx T'_{sys}$. With the control software of the KRT3 RadioUniversePro the easiest way to measure the background radiation is to do a cross scan over a weak source that barely stands out from the background.

For our purposes we chose the moon as it was at an elevation of 30°52 min and the recommended range of elevation is between 30 and 40 degrees of elevation.

2.4. Determination of the angular resolution of the KRT3

The expected half-width is $q = 1.02\lambda/D = \dots$ with the wavelength $\lambda = \dots$ and the antenna diameter $D = 3$ m. To determine the angular resolution the radio telescope is moved step wise over the sun and the antenna temperatures are measured.

The concrete settings in RadioUniversePro are a cross-scan in the Total Power Plots-tab with:

- Reference frame: Horizon
- Span [deg]: 20
- Step [deg]: 1
- Average [s]: 3
- Devices: BBCs
- Save as FITS

2.5. Determination of the Sensitivity of the instrument

2.5.1. Theoretical sensitivity

To determine the theoretical sensitivity of the instrument, the weakest detectable flux density ΔS_ν for a given system temperature T'_{sys} can be given by the radiometer equation:

$$\Delta S_\nu = \frac{2k C_s T'_{sys}}{A_e \sqrt{\Delta\nu\tau}} \quad (9)$$

where $C_s \approx 2$ is a dimensionless instrumental constant and τ the integration time. With Equation 9 the limiting sensitivity in a single channel with $\Delta\nu = 61$ kHz and an integration time of $\tau = 1$ s can be calculated to:

$$\Delta S_{\nu,\text{left}} = 6,248 \cdot 10^{-24} \frac{\text{W}}{\text{m}^2\text{Hz}} = 624,92 \text{ Jy} \quad (10)$$

$$\Delta S_{\nu,\text{right}} = 6,82 \cdot 10^{-24} \frac{\text{W}}{\text{m}^2\text{Hz}} = 682,02 \text{ Jy} \quad (11)$$

With $T'_{\text{sys, left}} = 98.76 \pm 22.54$ K, $T'_{\text{sys, right}} = 107.80 \pm 31.65$ K, $A_e = 1.125\pi$, $\tau = 1$ s, $\Delta\nu = 61$ kHz.

The measurement is carried out with the radio source Vir A.

The necessary integration time for detecting Vir A with a signal-to-noise ratio of $S/N = 10$ in each channel is calculated, taking the flux density at $\nu = 1420,4$ MHz to be $S_\nu = 211 \text{ Jy} \pm 11 \text{ Jy}$ after Baars et al. (1977):

- Virgo A: Flux density = 211 ± 11 Jy, Radiogalaxy
- For 5 channels on each side ($\Delta\nu = 5 \cdot 61\text{kHz} = 305\text{kHz}$)
- $S/N = 10$
- $\tau = \left(\frac{2k \cdot C_s \cdot T'_{\text{sys}}}{\Delta S_\nu \cdot A_e} \right)^2 \cdot \frac{\Delta\nu}{S/N}$
- $\tau_l = 4,12$ s
- $\tau_r = 5,22$ s

2.5.2. Determination of the uncertainty of radiant flux measurements

The radio signals from space recieved on earth are weak. This can be illustrated by the thought experiment of recieving the signal of a mobile radio on the moon transmitting with 2 W, 900 MHz and a FWHM of 80 kHz, $\sigma = 34$ kHz being recieved on earth. In this scenario the flux density is:

$$S_\nu = \frac{P}{4\pi d^2 \sqrt{2\pi\sigma}} = 1263,16 \text{ Jy} \quad (12)$$

When compared with the flux density of Cas A at 900 MHz ($1420 \text{ Jy} \pm 70 \text{ Jy}$), it's surprisingly high.

To calculate the flux density of Cas A, we assume the spectral response is a simple power law of $S_\nu \propto \nu^\alpha$, $\alpha = -0,7$: $\nu = 1420.4 \text{ MHz}$: $S_{\nu, \text{Cas A}} = 1420 \pm 70 \text{ Jy}$.

Using the previous equation $S_\nu = f(T_A, A_e)$, the antenna temperature T_A for Cas A can be calculated:

$$c = \frac{1420,4^{-0,7} \text{ MHz}}{1420 \text{ Jy}} \quad (13)$$

$$S_{\nu, \text{Cas A}} = \frac{900^{-0,7}}{c} \text{ Jy} = 1954,37 \text{ Jy} \quad (14)$$

$$T_A = \frac{S_\nu A_e}{2k_B} = 1,82 \text{ K} \quad (\text{für } \nu = 1420.4 \text{ MHz}) \quad (15)$$

To measure the flux density means to measure the source (On) and then to do a measurements without the source (Off). As we need to separate the background from the antenna temperature, a so-called Beamswitch measurement needs to carried out in which the position of the source and a reference position next to the source are measured alternately.

From this On-Off measurement the temperature of the source T_s can be determined the following way:

$$T_A(ON) - T_A(OFF) = (T_s + T'_{sys}) - T'_{sys} \quad (16)$$

The source Vir A was chosen to be observed in a grid pattern on the sky where the KRT3 takes measurements line by line at differnt positions. This means different columns are scanned to assemble the grid pattern into a map in the shape of pixel = rows $\times$ columns. The reason Vir A was chosen and not one of the brighter sources, was that Vir A was the only source that fit the parameters of an elevation between 30° and 60° .

The BBCs 3 and 11 were chosen as they showed the smallest RFI influence even though they show the H1 line.

The measurements were carried out in the tab Mapping with the following settings:

- Width: 10 deg; Height: 10 deg
- Delta: 0.5 HPBW
- Average: 10s

The mapping process took 28:35.4s according to the software and 29:27.8s according to our own timer.

2.6. Observation of neutral hydrogen in the Milky Way

Hydrogen can be detected with the help of the 21 cm line. The line exists because of a transition into the ground state of the neutral atom. Because Hydrogen is the most abundant element in the atmosphere and because radio waves aren't blocked by the interstellar dust, it's so practical to use the 21 cm line. Due to the doppler effect impacting the observed frequencies, the radial velocities of gas along a line of sight can be determined.

One of the approximations for doing exactly that, is the assumption that the hydrogen is distributed in the plane on circular orbits around the galactic centre. This approximation leads to the following orbital velocity:

$$v(R)^2 = \frac{GM_R}{R} \quad (17)$$

With M_R being the mass of the Milky Way within the radius R and the gravitational constant $G = 6,67 \cdot 10^{-11} \text{ m}^3/\text{kg}/\text{s}^2$.

As we observe from a point in the same system we want to measure, we need to think about how to actually measure the variable R . We observe along a line of sight defined through the relative angle l with respect to the line connecting the sun and the galactic centre (see Figure 1). This means our calculation for R , the distance to the point with the highest radial velocity, is $R = R_0 \sin(l)$ with $R_0 = 8,5 \text{ kpc}$ being the distance of the sun from the galactic centre.

The orbital velocity at the tangential point is $v = \omega R$ with ω being the angular velocity at distance R .

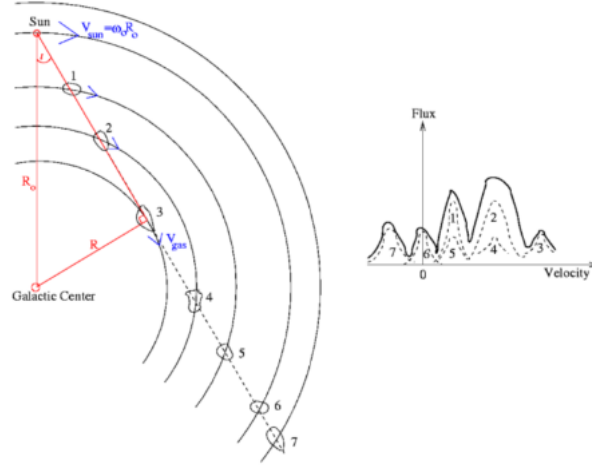


Figure 1: A schematic representation of the rotation of H1 clouds around the galactic centre together with the resulting line profile to the right [?].

This leads to the radial velocity between the tangent point and the sun:

$$v_{\text{rad}} = \omega(R)R - \omega_0 R_0 \sin(l) \quad (18)$$

$$= v(R) - \omega_0 R \quad (19)$$

The measurement carried out is an On-Off measurement at a point on the galactic plane. The location of the galactic centre is at $l = 0$, $b = 0$ with l being the galactic longitude and b being the galactic latitude. The spectra were taken with constant b and varied l in the range of 10° to 90° .

For that, the telescope is aligned to the galactic plane using the command line in RadioUniversePro. The USBs 3 and 11 in the BBC Tools tab were selected and the following settings were used:

- Repetitions: 10
- Average [s]: 10
- Elev cutoff [deg]: 60
- Distance [beam]: 4
- galoff
- Aquire OnOff Spectra
- Save as FITS

3. Results

3.1. Determination of the system temperature

The system temperature was determined from the calibration measurements using the relation

$$T'_{\text{sys}} = T_{\text{cal}} \frac{P_{\text{off}}}{P_{\text{cal}} - P_{\text{off}}},$$

where T_{cal} is the calibration temperature and P_{off} and P_{cal} are the received powers with the noise source switched off and on, respectively. The resulting values were averaged separately for the left circular polarization (BBC 1–8) and the right circular polarization (BBC 9–16), yielding

$$T'_{\text{sys, left}} = 98.76 \pm 22.54 \text{ K}, \quad T'_{\text{sys, right}} = 107.80 \pm 31.65 \text{ K}.$$

Using these system temperatures, the antenna temperature of the Sun was then calculated from the corresponding calibration data, giving

$$T_{\text{sun, left}} = 1237.79 \pm 89.62 \text{ K}, \quad T_{\text{sun, right}} = 1302.59 \pm 119.16 \text{ K}.$$

The larger uncertainty in the right polarization reflects the larger scatter of the corresponding measurements. With ?? and the current measured flux densities provided by the SWC of the Australian Government ($S_\nu \approx 76 \text{ sfu}$) the expected antenna temperature was estimated to be

$$T_A = 972.753 \text{ K}.$$

This value is in reasonable agreement with the measured temperature, confirming that the calibration procedure yields a consistent estimate of the solar signal.

3.2. Background Radiation

In order to convert the raw TPI signal into the actual background temperature, the measured data was normalized using the mean off-source power and the effective system temperature according to

$$T_{h,n} = \frac{\text{TPI} \cdot T'_{\text{sys}}}{\bar{P}_{\text{off}}}. \quad (20)$$

Here, T'_{sys} is the effective system temperature and $\bar{P}_{\text{off}}$ is the mean reference power measured during the off-source observation. This normalization removes the dependence on the receiver gain and instantaneous power level and rescales the measured TPI signal into an effective temperature. The resulting background curve (see Figure 2) therefore represents the true elevation-dependent contribution of the environment to the antenna temperature, which must be corrected for in order to recover the source signal reliably.

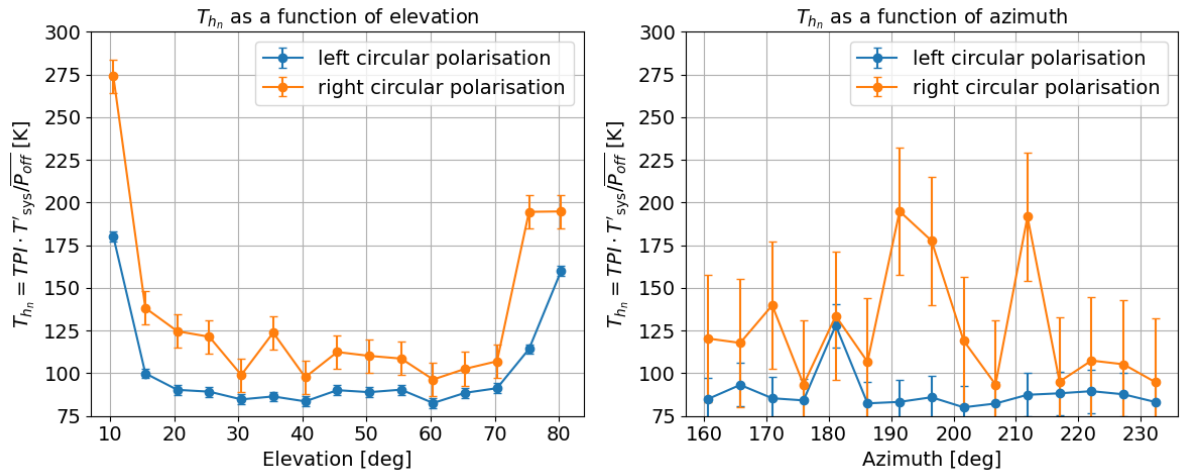


Figure 2: Normalised Background Temperature as a function elevation (left figure) or azimuth (right figure)

The background temperature T_h is not constant, but varies significantly with elevation. The background plot shows a clear increase toward lower elevations, indicating that the antenna receives additional radiation from the atmosphere and surrounding environment when observing near the horizon. At higher elevations, the environmental contribution decreases and T_h approaches a lower value set mainly by sky brightness and instrumental noise.

The background temperature also varies with azimuth, showing that the local environment contributes differently depending on the telescope pointing direction. This is consistent with differing amounts of ground spillover and thermal emission from nearby structures in different azimuthal directions.

The uncertainty is significantly larger for the right-hand polarization than for the left-hand polarization, especially as a function of azimuth. This suggests that the right channel is more strongly affected by instrumental systematics and environmental variations. The smaller uncertainties in the left polarization indicate a more stable response and a more reliable background determination.

Using the extrapolated course of T'_h (see Figure 3) up to an elevation of 0° , it becomes clear that the background temperature rises strongly at low elevations. A comparison with the Earth's effective radiation temperature, $T_{\text{Earth}} \approx 252\text{ K}$, shows that this increase is of a comparable magnitude and therefore reflects the dominance of atmospheric and ground emission at low elevation.

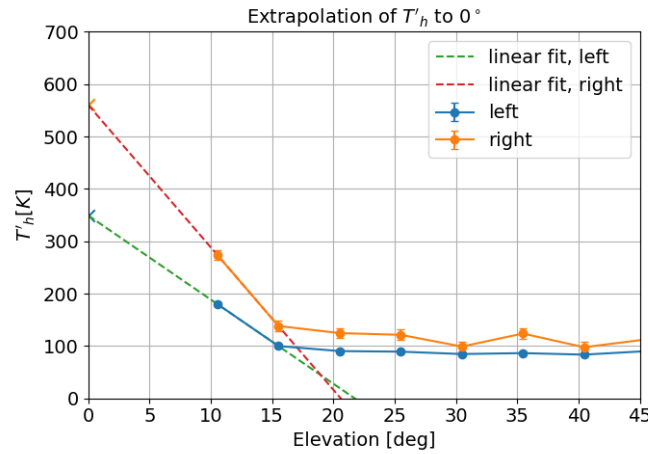


Figure 3: The extrapolated course of T'_h up to an elevation of 0° (for left and right hand polarisation)

3.3. Determination of the Angular Resolution of the KRT3

The angular resolution of the KRT3 was determined by scanning the Sun in longitude and latitude. The measured signal was normalized to the expected solar antenna temperature $T_{A,\text{Sun}}$. The resulting beam profiles were fitted with Gaussian functions, as shown in Figures 4 and 5. The angular resolution was obtained from the full width at half maximum (FWHM) of the fitted Gaussian profiles.

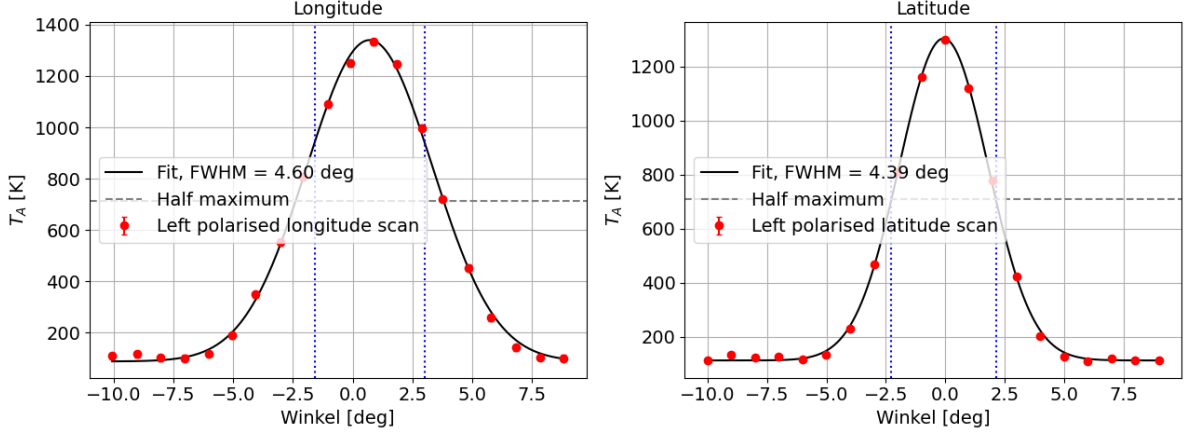


Figure 4: Gaussian fit of the

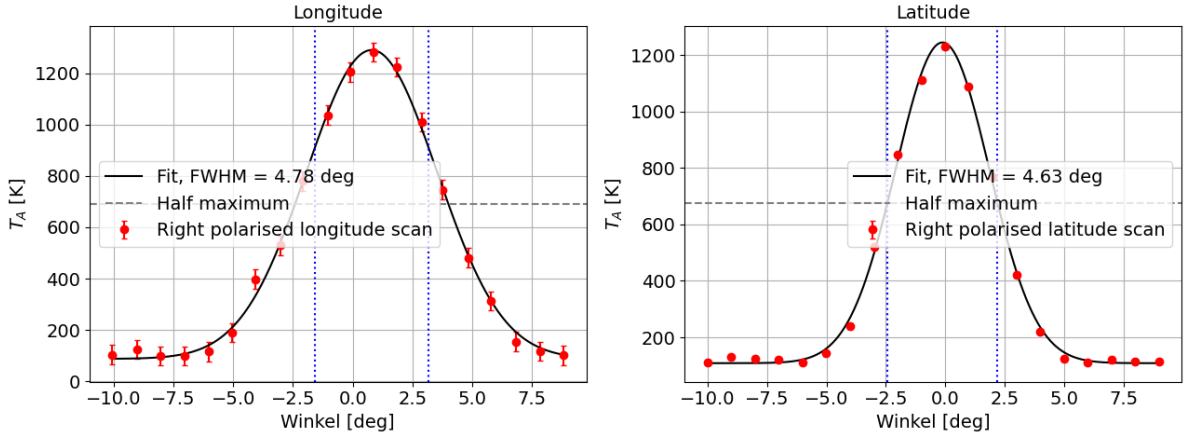


Figure 5: Gaussian fit of the

The resolution was determined separately for the longitude and latitude scans and for both circular polarization channels. In the longitude scans, the measured angular offset does not correspond directly to a great-circle distance on the sky. Therefore, the angular width was corrected using the factor $\cos(\text{ELV})$, where ELV is the elevation of the Sun ($\text{ELV} = 42^\circ$):

$$\theta_{\text{lon}} = \text{FWHM}_{\text{lon}} \cos(\text{ELV}).$$

For the latitude scans, no such correction is required because the scan follows a great circle approximately directly in elevation.

The theoretical angular resolution of the KRT3 is estimated using

$$\theta_{\text{theo}} \approx 1.02 \frac{\lambda}{D},$$

where $D = 3 \text{ m}$ is the antenna diameter and $\lambda = 21.1 \text{ cm} = 0.211 \text{ m}$ is the observing wavelength. This gives

$$\theta_{\text{theo}} \approx 1.02 \frac{0.211}{3} \approx 0.0717 \text{ rad} \approx 4.11^\circ.$$

The measured resolutions are slightly larger than this value for all scans. For the longitude scan, the measured values are

$$\theta_{\text{lon, left}} = 4.60^\circ, \quad \theta_{\text{lon, right}} = 4.78^\circ,$$

which are larger than the theoretical value by approximately 11.9% and 16.3%, respectively. For the latitude scan, the measured resolutions are

$$\theta_{\text{lat,left}} = 4.39^\circ, \quad \theta_{\text{lat,right}} = 4.63^\circ,$$

corresponding to deviations of approximately 6.8% and 12.7%.

The measured values are therefore in reasonable agreement with the theoretical resolution. The small systematic increase can be explained by pointing uncertainties, limited sampling, atmospheric fluctuations, imperfections in the Gaussian fit, and the finite angular diameter of the Sun. The right-circular polarization gives slightly larger values than the left-circular polarization, which is consistent with its larger measurement uncertainties and less stable receiver response.

3.4. Determination of the Sensitivity of the instrument

We used the software SAOImageDS9 for the evaluation of the measured map Figure 6. The mean of the map was calculated using a selected region of 35 pixels (see Figure 7) to be 1455.83 ± 212.206 . The map was normalised to the system temperature T'_{sys} which is then the $T_A(\text{OFF})$ from Equation 16. This equation is then used to calculate the antenna temperature of the source $T_s = 1612,17 \text{ K}$.

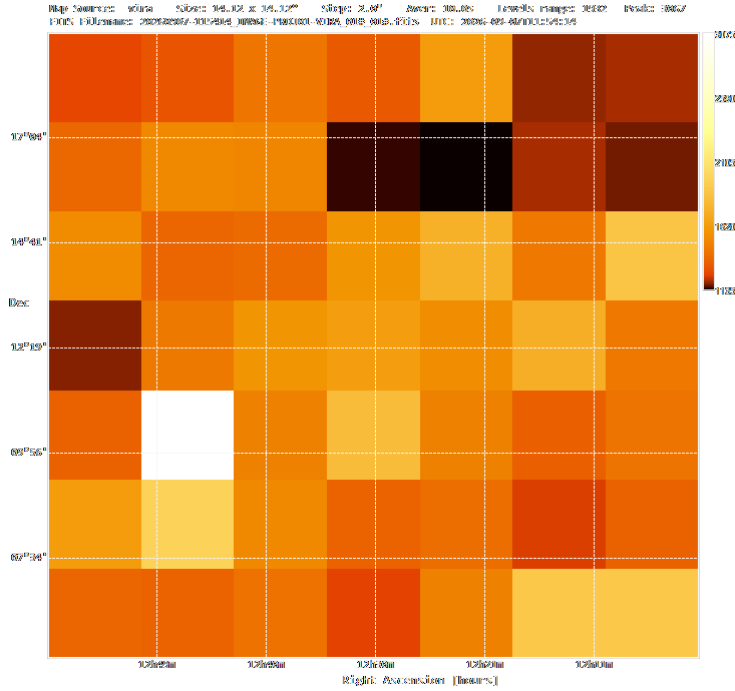


Figure 6: Map of the galaxy Virgo A

3.5. Observation of neutral hydrogen in the Milky Way

4. Discussion

Regarding the angular resolution, the noise is much larger than expected, especially when compared to the theoretical sensitivity of 600 Jy.

5. Summary

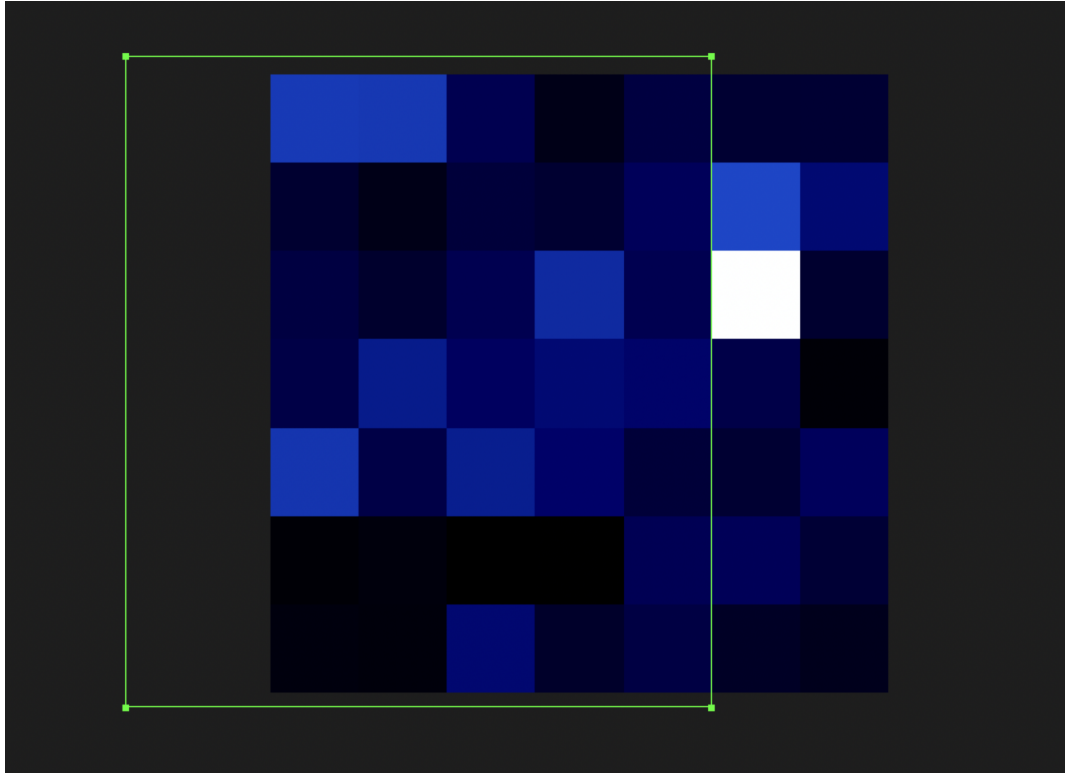


Figure 7: The selected region for the mean of the map, avoiding the probable disturbance.

References

- [1] H. Schmidt, "Normalengleichung." https://www.mathematik.uni-ulm.de/stochastik/lehre/ws05_06/statistik2old/skript/node21.html. Accessed: (15/03/2026).

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A. H_1 spectra

